

ASSIGNMENT 01

Q1. Consider 2-D Array: `int A[4][5];`
base address: 1000 size of int = 2 bytes.
Calculate memory address of following elements using both Row-major order and Column-major order.

Row-major: $(A[i][j]) = \text{lo} + [(i \cdot n) + j] \times w$ $m=4, n=5$

Column-major: $(A[i][j]) = \text{lo} + [(j \cdot m) + i] \times w$

• 1) $A[1][3]$

$$\text{row major} = 1000 + [(1 \times 5) + 3] \times 2 = 1016$$

$$\text{column major} = 1000 + [(3 \times 4) + 1] \times 2 = 1026$$

• 2) $A[2][4]$

$$\text{row major} = 1000 + [(2 \times 5) + 4] \times 2 = 1028$$

$$\text{column major} = 1000 + [(4 \times 4) + 2] \times 2 = 1036$$

• 3) $A[3][1]$

$$\text{row major} = 1000 + [(3 \times 5) + 1] \times 2 = 1032$$

$$\text{column major} = 1000 + [(1 \times 4) + 3] \times 2 = 1014$$

• 4) $A[0][2]$

$$\text{row major} = 1000 + [(0 \times 5) + 2] \times 2 = 1004$$

$$\text{column major} = 1000 + [(2 \times 4) + 0] \times 2 = 1016$$

• 5) $A[2][0]$

$$\text{row major} = 1000 + [(2 \times 5) + 0] \times 2 = 1020$$

$$\text{column major} = 1000 + [(0 \times 4) + 2] \times 2 = 1004$$

Sparse matrix:

perform following task:

- i) determine whether the given matrix is a sparse matrix
justify your answer.

	<u>0</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>
<u>0</u>	0	0	5	0	8
<u>1</u>	7	0	0	0	0
<u>2</u>	0	0	0	6	0
<u>3</u>	9	0	4	0	2

→ total elements : $4 \times 5 = 20$

total non-zero elements : 7

Zero elements : 13

since non-zero elements are more than zero elements, the given matrix is a sparse matrix.

ii) convert the given matrix into 3-tuple representation

row col = value

4	5	7
0	2	5
0	4	8
1	0	7
2	3	6
3	0	9
3	2	4
3	4	2

iii) calculate memory required

(a) normal representation

$$20 \times 2 = 40 \text{ bytes.}$$

(b) 3-tuple representation

row $\rightarrow 7+1=8$ (each row $\rightarrow 3$ integers)

$$\text{total : } 8 \times 3 = 24$$

$$\therefore \text{memory } 24 \times 2 = 48$$

bytes

memory:

$$= 24 \times 2$$

$$= 48 \text{ bytes.}$$

iv) Assume size of an integer is 2 bytes.

normal matrix:

$$20 \times 2 = 40 \text{ bytes.}$$

3-tuple representation:

$$8 \times 3 \times 2 = 48 \text{ bytes.}$$

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